THEORIES FOR THE BELL–KOCHEN–SPECKER THEOREM

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Abstract. We give a countable theory of real-valued $L_{\omega_1,\omega}$ whose standard models are the lattices of closed subspaces of separable Hilbert spaces equipped with quantum measures, and in which Gleason's theorem holds. No model of this theory, standard or not, carries a measure that takes only the values 0 and 1; thus the impossibility of hidden variables in the sense of Bell, Kochen and Specker follows from a countable theory.

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Introduction

In this paper, the lattice of closed subspaces of a separable complex Hilbert space, placed beside the Hilbert space and carrying a measure, becomes a standard model of a countable theory T of real-valued $L_{\omega_1,\omega}$, and Gleason's theorem and the Bell–Kochen–Specker theorem become statements about the models of T. The logic is the real-valued infinitary logic of the first author [Aub], which extends Eagle's infinitary $[0,1]$-valued logic [Eag14] and Keisler's real-valued logic [Kei23], and in which Keisler's model existence theorem holds. The vocabulary has a Hilbert sort, a Solèr sort for the lattice, a sort of square-summable sequences of vectors, and a real-valued predicate μ on the Solèr sort. A model is *standard* when its two vector sorts are the closed unit balls of complex Hilbert spaces (see Definition 5.2). For an element a of the Solèr sort, S_a is the closed subspace spanned by the vectors that belong to a (see Section 5.2), and P_a is the orthogonal projection onto S_a.

Theorem A (Gleason's theorem in real-valued $L_{\omega_1,\omega}$). *There is a countable theory T of real-valued $L_{\omega_1,\omega}$ with the following two properties.*

(1) In every standard model of T, the Hilbert sort is the closed unit ball of a separable infinite-dimensional complex Hilbert space H, the map $a \mapsto S_a$ is an isomorphism of ortholattices from the Solèr sort onto the lattice $L(H)$ of closed subspaces of H, and there is a unique density operator ρ on H such that $\mu(a) = \mathrm{Tr}(\rho P_a)$ for every element a of the Solèr sort.

(2) Conversely, for every separable infinite-dimensional complex Hilbert space H and every density operator ρ on H, there is a standard model of T whose Hilbert sort is the closed unit ball of H, whose Solèr sort is $L(H)$, and in which $\mu(a) = \mathrm{Tr}(\rho P_a)$ for every $a \in L(H)$.

Theorem B (The range of the measure). *Let T be the theory of Theorem A. In every standard model of T, the range of μ is the whole interval $[0,1]$. Consequently, for every Borel set $B \subseteq \mathbb{R}$ that meets $[0,1]$, the theory T together with the axiom*

$$\sup_a \mathbf{1}_B(\mu(a)) = 0,$$

which says that no value of μ lies in B, has no standard models; here $\mathbf{1}_B(\mu(a))$ is the formula of Lemma 5.1, whose value at a is 1 when $\mu(a) \in B$ and 0 otherwise.

Theorem C (The Bell–Kochen–Specker theorem in real-valued $L_{\omega_1,\omega}$). *The theory T of Theorem A, together with the axiom $\min(\mu(a), 1-\mu(a)) = 0$, which says that μ takes only the values 0 and 1, has no models, standard or not.*

A hidden-variable account of quantum mechanics would give every yes–no question about a system, that is, every closed subspace, a definite answer; the assignment of answers would be a measure with values 0 and 1. In standard models, the axioms of Theorem B exclude every measure whose range omits a point of $[0,1]$, since a single point is a Borel set; in particular, they exclude every two-valued measure. Two-valued measures are excluded from every model of T by Theorem C, including the countable models given by the downward Löwenheim–Skolem theorem, whose vector sorts are not complete; besides an orthonormal basis, its proof uses only the eighteen vectors of the configuration of Cabello, Estebaranz and García-Alcaine [CEGA96]. Thus the impossibility of hidden variables [Bel66, KS67] already follows from a countable theory of real-valued $L_{\omega_1,\omega}$, whose models need not be complete. Standardness cannot be dropped from Theorem B, since in a countable model the range of μ is countable and therefore omits intervals, nor from Theorem A(1), since the Solèr sort of a countable model is countable and therefore not of the form $L(H)$ (see Remark 10.2).

Roadmap. In Sections 1–3 we recall lattices, hermitian spaces and projective geometry, and the characterization of the lattices $L(H)$ due to Piron, Amemiya and Araki, and Solèr; the ortholattices that satisfy every condition of that characterization except completeness are the *pre-Solèr lattices*. Readers familiar with this material may skip these sections. In Section 4, which is optional, we determine what a pre-Solèr lattice lacks in order to be some $L(H)$; the proofs of Theorems A, B and C do not depend on it. In Sections 5 and 6 we

present the logic, the vocabulary and the axioms. In Section 7 we show that, without the sequence sort, the Solèr sort of a standard model is a pre-Solèr lattice whose σ-additive states Gleason's theorem still describes; we prove Theorems A and B in Section 8 and Theorem C in Section 9. The closing remarks of Section 10 are optional as well.

1. Lattices

Birkhoff and von Neumann [BvN36] observed that the experimentally verifiable propositions about a quantum system correspond to the closed subspaces of a Hilbert space, and that these form a lattice in which the distributive law fails; the logic of quantum mechanics is therefore not that of a Boolean algebra. They proposed the weaker modular law in its place, which holds in finite dimensions but fails for the closed subspaces of an infinite-dimensional Hilbert space. We begin with the lattice-theoretic terminology that this observation calls for.

Definition 1.1. A *partially ordered set* is a set with a reflexive, antisymmetric and transitive relation $\leq$. For a subset S of a partially ordered set, an *upper bound* of S is an element b with $a \leq b$ for all $a \in S$, and the *join* $\bigvee S$ of S is its least upper bound, when it exists; lower bounds and the *meet* $\bigwedge S$ are defined dually. A *lattice* is a partially ordered set in which every pair of elements a, b has a join $a \vee b$ and a meet $a \wedge b$. A lattice is *bounded* when it has a least element 0 and a greatest element 1, and *complete* when every subset has a join.

In a lattice, the relation $a \leq b$ holds if and only if $a \wedge b = a$; moreover, a lattice is complete if and only if every subset has a meet, since the meet of a subset is the join of its lower bounds.

Definition 1.2. An *orthocomplementation* on a bounded lattice Λ is a map $a \mapsto a^\perp$ from Λ to Λ such that, for all $a, b \in \Lambda$, the inequality $a \leq b$ implies $b^\perp \leq a^\perp$, and $a^{\perp\perp} = a$, $a \wedge a^\perp = 0$ and $a \vee a^\perp = 1$. A bounded lattice with an orthocomplementation is an *ortholattice*. Two elements a, b of an ortholattice are *orthogonal*, written $a \perp b$, when $a \leq b^\perp$. An ortholattice is *orthomodular* when, for all a, b, the inequality $a \leq b$ implies $b = a \vee (b \wedge a^\perp)$.

In $L(H)$, orthomodularity says that every closed subspace N containing a closed subspace M is the orthogonal sum of M and $N \cap M^\perp$. In an ortholattice, the De Morgan law $(\bigvee S)^\perp = \bigwedge\{a^\perp : a \in S\}$ holds whenever the join exists.

Definition 1.3. An *atom* of a bounded lattice is an element $p \neq 0$ such that $b \leq p$ implies $b = 0$ or $b = p$. A bounded lattice is *atomistic* when every element a is the join of the atoms below it (this join being required to exist). It has the *covering property* when, for every atom p and every element a with $p \not\leq a$, no element lies strictly between a and $a \vee p$. It is *irreducible* when, for any two distinct atoms p and q, some atom r distinct from p and q lies below $p \vee q$.

For complete atomistic lattices with the covering property, irreducibility in this sense is equivalent to the absence of a decomposition as a product of two lattices with more than one element (see Maeda and Maeda [MM70]); the definition above is the one that the coordinatization theorem of Section 3 requires.

Definition 1.4. A *sub-ortholattice* of an ortholattice Λ is a subset that contains 0 and 1 and is closed under the operations $\vee$, $\wedge$ and $^\perp$ of Λ. An *embedding* of ortholattices is an injective map that preserves $\vee$, $\wedge$, $^\perp$, 0 and 1.

We shall also need a notion of density that concerns the order alone; no topology is involved. Let Γ be a complete lattice and Λ a subset of Γ. We say that Λ is *join-dense* in Γ when every $g \in \Gamma$ satisfies $g = \bigvee\{a \in \Lambda : a \leq g\}$, the join being computed in Γ, and that Λ is *meet-dense* in Γ when every $g \in \Gamma$ satisfies $g = \bigwedge\{a \in \Lambda : a \geq g\}$. For instance, $\mathbb{Q}$ is join-dense and meet-dense in the extended real line $[-\infty, \infty]$. For a partially ordered set Λ and $X \subseteq \Lambda$, let X^u and X^l be the sets of upper and lower bounds of X in Λ; a *cut* is a subset

X with $X = (X^u)^l$. The cuts, ordered by inclusion, form the *MacNeille completion* $\mathrm{DM}(\Lambda)$ of Λ [Mac37] (also called the Dedekind–MacNeille completion, which explains the notation), a complete lattice into which $a \mapsto \{b \in \Lambda : b \leq a\}$ embeds Λ, preserving all joins and meets that exist in Λ.

Proposition 1.5 (Banaschewski [Ban56], Schmidt [Sch56]). *Let Λ be a partially ordered set, and let Γ be a complete lattice that contains Λ as a join-dense and meet-dense subset, its order extending that of Λ. For $g \in \Gamma$, put $\Lambda_{\leq g} = \{a \in \Lambda : a \leq g\}$. Then $g \mapsto \Lambda_{\leq g}$ is an isomorphism of Γ onto $\mathrm{DM}(\Lambda)$ that restricts to the canonical embedding on Λ. If Λ is an ortholattice, an orthocomplementation of Γ that extends that of Λ is unique, and it is given by $g^\perp = \bigwedge\{a^\perp : a \in \Lambda,\ a \leq g\}$.*

Proof. By join-density, we have $g = \bigvee \Lambda_{\leq g}$; hence the upper bounds of $\Lambda_{\leq g}$ in Λ are the elements $b \in \Lambda$ with $b \geq g$. By meet-density, their common lower bounds are the elements $a \in \Lambda$ with $a \leq g$, and $\Lambda_{\leq g}$ is a cut. Join-density also shows that $g \leq g'$ if and only if $\Lambda_{\leq g} \subseteq \Lambda_{\leq g'}$. If X is a cut and $g = \bigvee X$ in Γ, then X is contained in $\Lambda_{\leq g}$; conversely, an element $a \in \Lambda$ with $a \leq g$ lies below every $b \in X^u$, since each such b is an upper bound of X in Γ, and therefore $a \in X = (X^u)^l$. The formula for $g^\perp$ follows from $g = \bigvee \Lambda_{\leq g}$ and the De Morgan law. ☺

2. Hermitian spaces

The coordinatization theorem, which we state in Section 3, turns the atoms of a lattice into the one-dimensional subspaces of a vector space over a division ring, and orthocomplementation into orthogonality with respect to a form. In this section we collect the facts about such spaces that we need; the decisive one is that orthomodularity of the lattice of orthogonally closed subspaces is a property of the space itself (see Proposition 2.8).

Definition 2.1. A *division ring* is a ring with $1 \neq 0$ in which every nonzero element is invertible; multiplication need not be commutative. An *involution* of a division ring Δ is a map $\alpha \mapsto \alpha^*$ such that $(\alpha + \beta)^* = \alpha^* + \beta^*$, $(\alpha\beta)^* = \beta^*\alpha^*$ and $\alpha^{**} = \alpha$ for all $\alpha, \beta \in \Delta$. Let E be a left vector space over Δ. A *hermitian form* on E is a map $\langle \cdot \mid \cdot \rangle \colon E \times E \to \Delta$ such that $\langle \alpha x + \beta y \mid z \rangle = \alpha\langle x \mid z \rangle + \beta\langle y \mid z \rangle$ and $\langle y \mid x \rangle = \langle x \mid y \rangle^*$ for all $x, y, z \in E$ and $\alpha, \beta \in \Delta$; it is *anisotropic* when $\langle x \mid x \rangle = 0$ implies $x = 0$. A *hermitian space* is a left vector space over a division ring with an involution, together with an anisotropic hermitian form.

It follows that $\langle x \mid \alpha y \rangle = \langle x \mid y \rangle \alpha^*$. The bracket is linear in its first argument, as is customary in mathematics; in Dirac notation, which physicists use, it is linear in the second. Over $\mathbb{Q}$ with the identity involution, the form $\sum_n x_n y_n$ on finitely supported sequences is anisotropic, since a sum of rational squares vanishes only when every square does; nothing in the definition forces the scalars to be real or complex.

Definition 2.2. In a hermitian space E, two vectors x and y are *orthogonal*, written $x \perp y$, when $\langle x \mid y \rangle = 0$. For $S \subseteq E$, let $S^\perp$ be the set $\{x \in E : \langle x \mid y \rangle = 0 \text{ for all } y \in S\}$. A subspace M is *orthogonally closed* when $M^{\perp\perp} = M$.

Orthogonality is symmetric, since $\langle y \mid x \rangle = \langle x \mid y \rangle^*$.

Definition 2.3 (The lattice $\mathcal{C}(E)$). For a hermitian space E, we write $\mathcal{C}(E)$ for the set of orthogonally closed subspaces of E, ordered by inclusion.

Every set of the form $S^\perp$ is a subspace, and it is orthogonally closed, since $S^{\perp\perp\perp} = S^\perp$; intersections of orthogonally closed subspaces are orthogonally closed. Hence $\mathcal{C}(E)$ is a complete lattice with least element $\{0\}$ and greatest element E; the meet of a family $(M_j)_{j \in J}$ is $\bigcap_j M_j$, and its join is $\big(\sum_j M_j\big)^{\perp\perp}$, which may be strictly larger than $\sum_j M_j$. For

$M \in \mathcal{C}(E)$, anisotropy gives $M \cap M^\perp = \{0\}$, whence $M \vee M^\perp = (M^\perp \cap M)^\perp = E$; thus $\mathcal{C}(E)$, with $M \mapsto M^\perp$, is a complete ortholattice.

Definition 2.4. A hermitian space E is *orthomodular* when $M + M^\perp = E$ for every $M \in \mathcal{C}(E)$, where $M + M^\perp$ is the set of sums $v + y$ with $v \in M$ and $y \in M^\perp$.

The lattice join $M \vee M^\perp$ is always E; orthomodularity asks that the algebraic sum be E as well.

Definition 2.5. A *Hilbert space* is a hermitian space H over the real numbers $\mathbb{R}$, the complex numbers $\mathbb{C}$ or the quaternions $\mathbb{H}$, with the usual involution (the identity, complex conjugation or quaternionic conjugation), such that $\langle x \mid x\rangle > 0$ for $x \neq 0$ and H is complete in the norm $\|x\| = \langle x \mid x\rangle^{1/2}$.

For a Hilbert space H, the orthogonally closed subspaces are exactly the norm-closed subspaces; thus $\mathcal{C}(H)$ is the lattice $L(H)$ of closed subspaces of H, and it is orthomodular by the projection theorem, which gives $M + M^\perp = H$ for every closed subspace M.

Lemma 2.6. *In a hermitian space E, every finite-dimensional subspace F is orthogonally closed and satisfies $F + F^\perp = E$.*

Proof. By anisotropy, the Gram–Schmidt process gives F an orthogonal basis $w_1, \dots, w_n$ with $\langle w_i \mid w_i\rangle \neq 0$ for $i = 1, \dots, n$. For $z \in E$, the vector $Pz = \sum_i \langle z \mid w_i\rangle \langle w_i \mid w_i\rangle^{-1} w_i$ lies in F, and $z - Pz$ lies in $F^\perp$; hence $F + F^\perp = E$. If $y \in F^{\perp\perp}$, the vector $y - Py$ lies in $F^\perp \cap F^{\perp\perp}$ and is therefore orthogonal to itself; by anisotropy it is 0, whence $y \in F$. ☺

Lemma 2.7. *Let E be a hermitian space over Δ. If $M \in \mathcal{C}(E)$ and $x \in E$, then $M + \Delta x \in \mathcal{C}(E)$.*

Proof. If $x \in M$, there is nothing to prove; we therefore assume that $x \notin M$. Put $W = M^\perp$, and let ξ_x be the functional $z \mapsto \langle z \mid x\rangle$ on W; it is nonzero, since $x \notin W^\perp = M$, and we may choose $z_0 \in W$ with $\xi_x(z_0) = 1$. Let y be an element of $(M + \Delta x)^{\perp\perp}$, and let ξ_y be the functional $z \mapsto \langle z \mid y\rangle$ on W. Since $W \cap x^\perp = (M + \Delta x)^\perp$, the functional ξ_y vanishes on the kernel of ξ_x; as $z - \xi_x(z) z_0$ lies in that kernel for every $z \in W$, we obtain $\xi_y(z) = \xi_x(z)\alpha$ with $\alpha = \xi_y(z_0)$. Because $\langle z \mid \alpha^* x\rangle = \langle z \mid x\rangle \alpha$, the vector $y - \alpha^* x$ is orthogonal to W and therefore lies in $W^\perp = M$. Thus $y \in M + \Delta x$. ☺

Proposition 2.8. *Let E be a hermitian space. The lattice $\mathcal{C}(E)$ is orthomodular if and only if E is an orthomodular space.*

Proof. Suppose first that E is orthomodular, and let $M \subseteq M'$ be elements of $\mathcal{C}(E)$. Every vector $z \in M'$ can be written as $v + y$ with $v \in M$ and $y \in M^\perp$; since $y = z - v$ lies in M', we have $M' = M + (M' \cap M^\perp) \subseteq M \vee (M' \wedge M^\perp) \subseteq M'$, which is the orthomodular law. Conversely, suppose that $\mathcal{C}(E)$ is orthomodular, let M be an element of $\mathcal{C}(E)$, and let x be a vector of $E \setminus M$. By Lemma 2.7, the subspace $M' = M + \Delta x$ belongs to $\mathcal{C}(E)$, and orthomodularity gives $M' = M \vee (M' \cap M^\perp)$. Since $M' \neq M$, the subspace $M' \cap M^\perp$ is nonzero; as it meets M only in 0, while M has codimension one in M', it is a one-dimensional subspace Δy. Then $M \subsetneq M + \Delta y \subseteq M'$, and the codimension count gives $M' = M + \Delta y$; hence $x \in M + M^\perp$. ☺

3. Projective geometry, Solèr lattices and pre-Solèr lattices

In this section we assume familiarity with Hilbert spaces, but none with projective geometry. Our purpose is to explain how an ortholattice with a few combinatorial properties can be represented by orthogonally closed subspaces of a vector space with a hermitian form, and how Solèr's condition forces the scalars to be real, complex or quaternionic.

3.1. **Points and lines.** In a complex Hilbert space H, the atoms of $L(H)$ are the one-dimensional subspaces $\mathbb{C}x$, with $x \neq 0$, called *rays*; the join of two distinct rays is the two-dimensional subspace, or *plane*, that they span. More generally, we call the one-dimensional subspaces of any vector space its *rays*. Which rays lie in which planes is information about the lattice alone (see Figure 1); the theorems below show that it determines the space up to the choice of scalars.

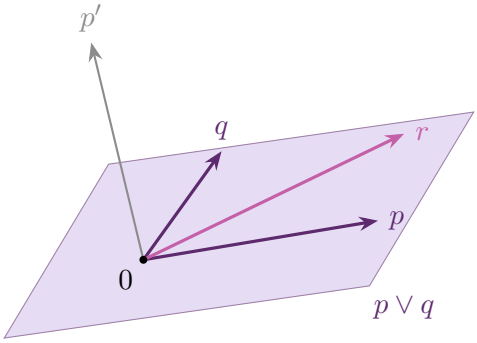


FIGURE 1. Atoms of $L(H)$ are rays. The join $p \vee q$ of two rays is the plane they span; the ray r in that plane is an atom below $p \vee q$, while the ray p' is not.

Definition 3.1. A *projective geometry* is a set of *points* together with a family of subsets, called *lines*, such that:

(PG1) any two distinct points lie on exactly one line;
(PG2) every line contains at least three points;
(PG3) (*the Veblen–Young axiom*) if a, b, a', b' are distinct points and the line through a and b meets the line through a' and b', then the line through a and a' meets the line through b and b'.

A set of points is a *projective subspace* when it contains the whole line through any two of its points; the projective subspace *generated* by a set of points is the intersection of all projective subspaces containing it, and a *projective plane* is the projective subspace generated by three points not on one line. Axiom (PG3) says that two lines lying in a common projective plane always meet (see Figure 2); a projective geometry has no parallel lines. The name comes from perspective drawing, in which parallel rails, projected onto the canvas, meet at the horizon.

The basic example comes from linear algebra. Let V be a left vector space over a division ring Δ, of dimension at least two. Take as points the rays of V, and as lines the sets of rays contained in a two-dimensional subspace. Axiom (PG1) holds because two distinct rays span a unique two-dimensional subspace, and axiom (PG2) because the two-dimensional subspace spanned by linearly independent vectors x and y contains the three distinct rays Δx, Δy and $\Delta(x + y)$. For (PG3), if the two-dimensional subspaces spanned by a, b and by a', b' have a ray in common, then a, b, a', b' lie in a subspace of dimension at most three, in which any two two-dimensional subspaces have a ray in common, by counting dimensions.

Points are *independent* when none of them lies in the projective subspace generated by the others; in the basic example, rays are independent exactly when nonzero vectors spanning them are linearly independent. The following theorem says that, as soon as there are four independent points, the basic example is the only one.

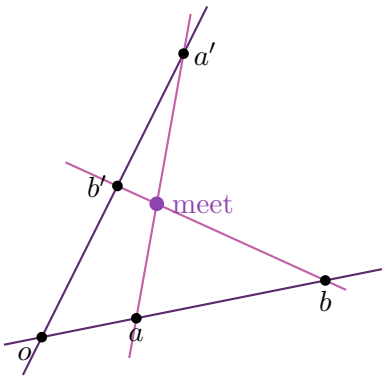


FIGURE 2. The Veblen–Young axiom. The lines ab and $a'b'$ meet at o; then the lines aa' and bb' meet as well.

Theorem 3.2 (Coordinatization of projective geometries; see Baer [Bae52]). *Every projective geometry that contains four independent points is isomorphic, by a bijection of points that maps lines onto lines, to the geometry of rays of a left vector space E over a division ring Δ. The pair (Δ, E) is unique up to a* semilinear isomorphism*: an additive bijection between the two vector spaces that is linear except that it moves scalars by an isomorphism between the two division rings.*

Two features of this theorem matter below: the division ring Δ is recovered from the geometry alone, and it can be any division ring, whether the rational numbers, a non-archimedean ordered field (one with infinitesimals, such as a field of formal Laurent series) or the quaternions; and the four independent points cannot be dispensed with. A projective plane, regarded as a projective geometry in its own right, need not come from a division ring: those that do are exactly the planes in which Desargues's theorem holds (if two triangles are in perspective from a point, the three intersection points of corresponding sides lie on one line), and the others, the *non-Desarguesian* planes, exist. In a geometry with four independent points, Desargues's theorem follows from the axioms.

3.2. **From lattices to geometries.** Let Λ be an atomistic lattice with the covering property. We take its atoms as the points of a geometry whose lines are the sets $\{r : r \text{ an atom with } r \leq p \vee q\}$, for distinct atoms p and q; in $L(H)$, these are the rays in the plane spanned by p and q. The covering property makes this geometry satisfy (PG1) and (PG3), and irreducibility in the sense of Definition 1.3 is exactly (PG2); see Maeda and Maeda [MM70, §§33–34] and Holland [Hol95, §3].

Orthocomplementation adds a relation between points: by the definition in Section 1, atoms p and q are orthogonal when $p \leq q^\perp$. In $L(H)$ this is orthogonality of rays, and it determines the inner product up to the ambiguity described at the end of Theorem 3.3 below. Consider the following properties of an ortholattice Λ:

(C) the lattice is complete;
(OM) it is orthomodular;
(A) it is atomistic;
(CP) it has the covering property;
(I) it is irreducible;
(D) it contains four mutually orthogonal atoms.

The following theorem, which goes back to Birkhoff and von Neumann [BvN36] in finite dimensions, says that orthogonality of atoms determines a form in the same way for every ortholattice with (OM), (A), (CP), (I) and (D). Condition (D) supplies the four independent points of Theorem 3.2, since mutually orthogonal atoms are independent.

Theorem 3.3 (Coordinatization; Baer [Bae52], Maeda and Maeda [MM70, §34], Holland [Hol95, §3]). *Let Λ be an ortholattice that satisfies (OM), (A), (CP), (I) and (D), but not necessarily (C). Then there are a division ring Δ with an involution, a hermitian space E over Δ, and a bijection $p \mapsto R_p$ from the atoms of Λ onto the rays of E, such that two atoms p and q are orthogonal exactly when $R_p \perp R_q$. The map*

$$\gamma\colon \Lambda \to \mathcal{C}(E), \qquad \gamma(a) = \bigcup\{R_p : p \text{ an atom with } p \le a\},$$

is an embedding of ortholattices whose image contains every finite-dimensional subspace of E and the orthogonal complement of each. The triple formed by Δ, E and the form is unique up to a semilinear isomorphism followed by a rescaling $\langle x \mid y\rangle' = \langle x \mid y\rangle\zeta$, where $\zeta^* = \zeta \neq 0$, with the new involution $\alpha \mapsto \zeta^{-1}\alpha^*\zeta$.

Proof of the statements about γ. The orthocomplementation of Λ induces a *polarity* on the geometry of its atoms: an inclusion-reversing bijection of the projective subspaces onto themselves, whose square is the identity, which sends each projective subspace to the set of points orthogonal to all its points. By Theorem 3.2, the atoms become the rays of a vector space E, and by Baer's theorem [Bae52], every polarity of this kind under which no point is orthogonal to itself is orthogonality for an anisotropic hermitian form on E. If two atoms lie below a, every atom on the line they determine lies below a; hence $\gamma(a)$ is a subspace. For an atom r, the following are equivalent: $r \le a^\perp$; $a \le r^\perp$; $p \le r^\perp$ for every atom $p \le a$ (because a is the join of these atoms, by (A)); $R_r \perp R_p$ for every atom $p \le a$. Thus $\gamma(a^\perp) = \gamma(a)^\perp$, and applying this twice gives $\gamma(a) = \gamma(a)^{\perp\perp}$. By (A), the relation $a \le b$ holds if and only if $\gamma(a) \subseteq \gamma(b)$; hence γ is injective. Meets are preserved because an atom lies below $a \wedge b$ exactly when it lies below a and b, and joins are preserved by the De Morgan law. If $x_1, \dots, x_k$ is a basis of a finite-dimensional subspace and p_i is the atom with $R_{p_i} = \Delta x_i$, the subspace is $\gamma(p_1 \vee \cdots \vee p_k)$, the image of a finite join, which exists in every lattice; its orthogonal complement is $\gamma\big((p_1 \vee \cdots \vee p_k)^\perp\big)$. ☻

Piron [Pir64] proved this representation for complete lattices; completeness is not needed for the embedding, and in Theorem 3.6 below it serves only to make γ surjective.

3.3. **Harmonic conjugates and Solèr's condition.** The coordinatization leaves the division ring Δ undetermined. Solèr's condition singles out the real, complex and quaternionic numbers, and it is phrased through a classical construction of projective geometry.

Let p and q be distinct points, write $p + q$ for the line through them (in the basic example, the rays of the plane spanned by p and q), and let c be a point of $p + q$ other than p and q. Two distinct lines in a common projective plane meet in exactly one point, by (PG3), and we denote that point by an intersection sign. Choose a point r not on $p + q$ and a point r' on $r + p$ other than r and p; put $c' = (c + r') \cap (q + r)$ and $c'' = (p + c') \cap (q + r')$. The point $h = (r + c'') \cap (p + q)$ does not depend on the choices of r and r', as the computation below shows; it is the *harmonic conjugate* of c with respect to p and q (see Figure 3). In a lattice satisfying (A), (CP) and (I), the same construction applies to atoms, with $\vee$ in place of $+$ and $\wedge$ in place of $\cap$; the relation "h is the harmonic conjugate of c with respect to p and q" is then expressed by a first-order formula in the language of lattices.

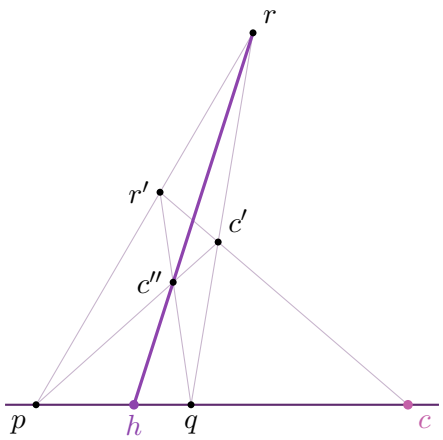


FIGURE 3. The harmonic conjugate h of c with respect to p and q.

In coordinates the construction is simple. Let u and v be vectors spanning p and q, scaled so that $c = \Delta(u + v)$, and let w be a vector spanning r, scaled so that $r' = \Delta(u + w)$, which is possible because r' lies on $r + p$ and differs from r and p. A direct computation gives $c' = \Delta(v - w)$, $c'' = \Delta(u - v + w)$ and $h = \Delta(u - v)$. All coefficients are ± 1, so the computation is valid over every division ring:

$$\text{the harmonic conjugate of } \Delta(u+v) \text{ with respect to } \Delta u \text{ and } \Delta v \text{ is } \Delta(u-v). \tag{1}$$

Now suppose that $p \perp q$; then $u \perp v$. The orthocomplement of c within the plane $p \vee q$, the ray $c^\perp \wedge (p \vee q)$, consists of the vectors $\alpha u + \beta v$ with $\langle \alpha u + \beta v \mid u + v \rangle = \alpha \langle u \mid u \rangle + \beta \langle v \mid v \rangle = 0$. It coincides with the harmonic conjugate $\Delta(u - v)$ exactly when $\langle u \mid u \rangle = \langle v \mid v \rangle$, that is, when u and v have the same "length" (see Figure 4). In characteristic 2, that is, when $1 + 1 = 0$ in Δ, the vectors $u + v$ and $u - v$ coincide, and the equality $\langle u \mid u \rangle = \langle v \mid v \rangle$ would make $u + v$ isotropic, since then $\langle u + v \mid u + v \rangle = (1 + 1)\langle u \mid u \rangle = 0$. The equality of the two rays therefore forces the characteristic to differ from 2.

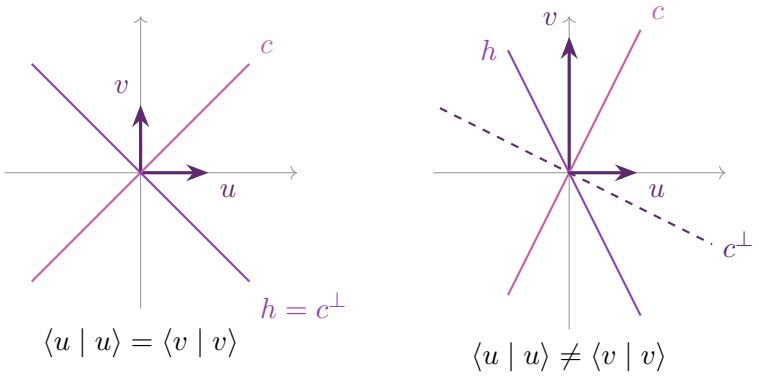


FIGURE 4. The ray $c = \Delta(u + v)$, its harmonic conjugate $h = \Delta(u - v)$, and its orthocomplement in the plane of u and v. They coincide when u and v have equal length (left), and differ otherwise (right).

Solèr's condition, in the form we use, asks for an infinite orthogonal sequence of atoms along which the orthocomplement and the harmonic conjugate agree at every step:

(S) There are mutually orthogonal atoms p_n, for $n \in \mathbb{N}$, and atoms $c_n \leq p_n \vee p_{n+1}$ with $c_n \neq p_n, p_{n+1}$, such that, for every $n \in \mathbb{N}$, the harmonic conjugate of c_n with respect to p_n and p_{n+1} is $c_n^\perp \wedge (p_n \vee p_{n+1})$.

This formulation of Solèr's condition through harmonic conjugates is due to Holland [Hol95, Theorems 3.7 and 4.1]; Morash's angle-bisecting axiom [Mor73], which asks for bisectors as in the left half of Figure 4, may be used in its place (see Holland [Hol95, §4]).

Lemma 3.4. *Let Λ be an ortholattice that satisfies (OM), (A), (CP), (I), (D) and (S), and let E be as in Theorem 3.3. After a rescaling of the form, the space E contains an infinite orthonormal sequence.*

Proof. Choose e_0 with $R_{p_0} = \Delta e_0$, and inductively choose e_{n+1} with $R_{p_{n+1}} = \Delta e_{n+1}$ and $R_{c_n} = \Delta(e_n + e_{n+1})$; this is possible because R_{c_n} is a ray of the plane spanned by e_n and e_{n+1}, distinct from Δe_n and Δe_{n+1}. By (1) and the computation that follows it, condition (S) gives $\langle e_n \mid e_n \rangle = \langle e_{n+1} \mid e_{n+1} \rangle$ for every $n \in \mathbb{N}$; call this common value ζ, which satisfies $\zeta^* = \zeta \neq 0$. The form $\langle x \mid y \rangle' = \langle x \mid y \rangle \zeta^{-1}$ is hermitian for the involution $\alpha \mapsto \zeta \alpha^* \zeta^{-1}$, has the same orthogonality relation, and satisfies $\langle e_n \mid e_n \rangle' = 1$ for every n. ☺

Definition 3.5. A *Solèr lattice* is an ortholattice that satisfies (C), (OM), (A), (CP), (I), (D) and (S). A *pre-Solèr lattice* is an ortholattice that satisfies (OM), (A), (CP), (I), (D) and (S), but not necessarily (C).

Theorem 3.6 (Piron, Amemiya and Araki, Solèr; in this form Holland [Hol95, Theorem 4.1]**).** *An ortholattice is a Solèr lattice if and only if it is isomorphic to $L(H)$ for some infinite-dimensional Hilbert space H over $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$.*

Sketch. Let Λ be a Solèr lattice, with γ and E as in Theorem 3.3. Given $M \in \mathcal{C}(E)$, completeness provides the join a of the atoms p with $R_p \subseteq M$, and $\gamma(a)^\perp = \gamma(a^\perp) = M^\perp$, whence $\gamma(a) = M$; thus γ is an isomorphism, $\mathcal{C}(E)$ is orthomodular, and E is an orthomodular space by Proposition 2.8. By Lemma 3.4, the space E contains, after a rescaling, an infinite orthonormal sequence, and Solèr's theorem [Sol95] (see also Prestel [Pre95]) shows that E is a Hilbert space over $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$. Conversely, $L(H)$ is complete and satisfies the other conditions; condition (S) holds with p_n the ray of e_n and c_n the ray of $e_n + e_{n+1}$, for an orthonormal sequence $(e_n)_{n \in \mathbb{N}}$. ☺

Keller [Kel80] constructed orthomodular hermitian spaces that are not Hilbert spaces; hence condition (S) cannot be omitted from Theorem 3.6.

4. What a pre-Solèr lattice is missing (optional)

A pre-Solèr lattice satisfies every condition that defines a Solèr lattice except (C). We determine what it lacks: some elements, which an order-theoretic completion supplies, and possibly the orthomodularity of that completion. Neither deficiency can be detected by a sentence of $L_{\omega_1,\omega}$ about the lattice alone (see Proposition 4.5 below); this is why the vocabulary of Section 5 has vector sorts. From now on, Λ is a pre-Solèr lattice; we fix a hermitian space E that Theorem 3.3 provides for Λ, and we identify Λ with its image $\gamma(\Lambda) \subseteq \mathcal{C}(E)$.

Theorem 4.1. *Let Λ be a pre-Solèr lattice, identified as above with its image in $\mathcal{C}(E)$. Then Λ is join-dense and meet-dense in $\mathcal{C}(E)$; consequently $\mathcal{C}(E)$, with $M \mapsto M^\perp$, is isomorphic as an ortholattice to the MacNeille completion of Λ.*

Proof. Every $M \in \mathcal{C}(E)$ is the join in $\mathcal{C}(E)$ of the rays it contains, which belong to Λ by Theorem 3.3. If $M^\perp = \{0\}$, then M is E, which belongs to Λ; otherwise M equals $M^{\perp\perp} =$

$\bigcap\{y^\perp : 0 \neq y \in M^\perp\}$, a meet of hyperplanes, each of which belongs to Λ. Proposition 1.5 gives both assertions. ☺

The elements of the MacNeille completion that do not come from Λ are therefore orthogonally closed subspaces M with $\dim M = \dim M^\perp = \infty$, since Λ contains every finite-dimensional subspace and the orthogonal complement of each. Adjoining them need not produce some $L(H)$, because the completion may fail to be orthomodular (see Example 4.3); the next theorem shows that orthomodularity of the completion is exactly what is needed.

Theorem 4.2. *For a pre-Solèr lattice Λ, the following are equivalent:*

(1) the MacNeille completion of Λ is orthomodular;
(2) the hermitian space E that Theorem 3.3 provides for Λ is orthomodular;
(3) Λ is isomorphic to a join-dense and meet-dense sub-ortholattice of $L(H)$ for some infinite-dimensional Hilbert space H over $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$.

When these conditions hold, E with its form suitably rescaled is a Hilbert space as in (3), and Λ contains every finite-dimensional subspace of E and the orthogonal complement of each. Moreover, Λ is isomorphic to $L(H)$ for such an H if and only if Λ is complete.

Proof. Conditions (1) and (2) are equivalent by Theorem 4.1 and Proposition 2.8. If (2) holds, Lemma 3.4 and Solèr's theorem show that E, with its form rescaled, is a Hilbert space; rescaling changes neither $\mathcal{C}(E)$ nor its orthocomplementation, and Theorem 4.1 gives (3). If (3) holds, Proposition 1.5 identifies $L(H)$, which is orthomodular, with the MacNeille completion of Λ, together with its orthocomplementation; this is (1). For the last assertion, a complete lattice coincides with its MacNeille completion, and $L(H)$ is complete. ☺

Example 4.3. Let E_0 be the space of finitely supported complex sequences, with the inner product $\langle x \mid y\rangle = \sum_n x_n\overline{y_n}$ of the space ℓ^2 of square-summable sequences, and let Λ_0 be the set of finite-dimensional subspaces of E_0 and their orthogonal complements in E_0. Then Λ_0 is a sub-ortholattice of $\mathcal{C}(E_0)$, since $F \vee G^\perp = (F^\perp \cap G)^\perp$ for finite-dimensional F and G. It is orthomodular, because every $M \in \Lambda_0$ satisfies $M + M^\perp = E_0$ by Lemma 2.6, and the argument of Proposition 2.8 applies. Since it contains every ray of E_0, it satisfies (A), (CP) and (I); conditions (D) and (S) hold with $p_n = \mathbb{C}e_n$ and $c_n = \mathbb{C}(e_n + e_{n+1})$, where $(e_n)_{n\in\mathbb{N}}$ is the sequence of standard unit vectors. Its MacNeille completion, $\mathcal{C}(E_0)$ by Theorem 4.1, is not orthomodular: by the Amemiya–Araki theorem [AA66], a complex inner product space whose lattice of orthogonally closed subspaces is orthomodular is complete, and E_0 is not. The lattice Λ_0 is thus an orthomodular lattice whose MacNeille completion is not orthomodular; Harding [Har91] studies such lattices.

Orthomodularity of the completion can be stated through the atoms of Λ alone. For a set $\mathcal{A}$ of atoms, let $\mathcal{A}^\perp$ be the set of atoms orthogonal to every element of $\mathcal{A}$, and call $\mathcal{A}$ *orthoclosed* when $\mathcal{A} = \mathcal{A}^{\perp\perp}$. Consider the following condition, named (MOM) for MacNeille orthomodularity:

(MOM) for every orthoclosed set $\mathcal{A}$ of atoms and every atom $p \notin \mathcal{A}$, some atom $q \in \mathcal{A}^\perp$ lies in $(\mathcal{A} \cup \{p\})^{\perp\perp}$.

Theorem 4.4. *The MacNeille completion of a pre-Solèr lattice is orthomodular if and only if the lattice satisfies (MOM).*

Proof. The map $\mathcal{A} \mapsto \bigcup\{R_p : p \in \mathcal{A}\}$ is a bijection from the orthoclosed sets of atoms onto $\mathcal{C}(E)$, carrying $\mathcal{A}^\perp$ to the orthogonal complement of the image of $\mathcal{A}$. If M is the image of $\mathcal{A}$ and $R_p = \Delta x$, then by Lemma 2.7 the set $(\mathcal{A} \cup \{p\})^{\perp\perp}$ corresponds to $M + \Delta x$. Condition (MOM) therefore says that $(M + \Delta x) \cap M^\perp \neq \{0\}$ whenever $M \in \mathcal{C}(E)$ and $x \notin M$. If

$y = v + \alpha x$ is nonzero, with $v \in M$ and $y \in M^\perp$, then the scalar α is nonzero, since $y \notin M$, and $x = \alpha^{-1}(y - v)$ lies in $M + M^\perp$; conversely, if $x = v + y$ with $v \in M$ and $y \in M^\perp$, then $y = x - v$ is a nonzero element of $(M + \Delta x) \cap M^\perp$. Thus (MOM) is equivalent to the orthomodularity of E, and Theorem 4.2 applies. ☺

When $\mathcal{A}$ corresponds to a finite-dimensional subspace, the requirement in (MOM) holds automatically, by Lemma 2.6; its content lies in the infinite orthoclosed sets of atoms, over which it quantifies.

Proposition 4.5. *Neither (C) nor (MOM) is equivalent, among pre-Solèr lattices, to a sentence of two-valued $L_{\omega_1,\omega}$ in the vocabulary of ortholattices expanded by constants for the atoms p_n and c_n of (S).*

Proof. Suppose that a sentence φ of $L_{\omega_1,\omega}$ were equivalent to (MOM) among pre-Solèr lattices. The axioms of ortholattices and the conditions (OM), (A), (CP), (I), (D) and (S) are expressed by countably many sentences of $L_{\omega_1,\omega}$ in this vocabulary; together with φ, they hold in $L(\ell^2)$. By the downward Löwenheim–Skolem theorem for a countable fragment, they hold in a countable structure Λ', which is a pre-Solèr lattice satisfying (MOM). By Theorems 4.4 and 4.2, the lattice Λ' contains every ray of an infinite-dimensional Hilbert space, and is therefore uncountable, a contradiction. The same argument applies to (C), using Theorem 3.6. ☺

5. The logic and the vocabulary

Completeness, the one property that separates the lattices $L(H)$ from the pre-Solèr lattices, cannot be expressed by sentences of $L_{\omega_1,\omega}$, two-valued or real-valued, about the lattice alone: by the downward Löwenheim–Skolem theorem, any countable set of such sentences that holds in some $L(H)$ also holds in a countable lattice, while every Solèr lattice is uncountable (for the details, see the optional Proposition 4.5). A Hilbert space can supply the missing joins: we place the lattice beside a Hilbert space and a space of sequences of vectors, and connect the three by symbols whose intended meanings are projection, membership and closed span.

5.1. **Real-valued $L_{\omega_1,\omega}$.** We use the real-valued infinitary logic of the first author [Aub], which extends both Eagle's infinitary $[0,1]$-valued logic [Eag14] and Keisler's real-valued logic [Kei23]; Keisler's model existence theorem extends to it. Of this logic we use only the following features:

- ♠ The vocabulary is countable and many-sorted; no metric is given a priori, and function symbols need not be continuous.
- ♠ Predicate symbols are real-valued, each with values in a fixed bounded interval; every formula therefore takes its values in a bounded interval, whose bound depends on the formula.
- ♠ Formulas are closed under continuous connectives, under the quantifiers sup and inf over each sort, and under countable suprema $\sup_n \varphi_n$ and infima $\inf_n \varphi_n$ of families of formulas that are uniformly bounded and have finitely many free variables in all.
- ♠ There is no equality symbol; as in [Kei23], a structure is *reduced* when any two elements that agree in every atomic formula with parameters are equal. Models are taken to be reduced.

A *condition* is an expression $\varphi = 0$ or $\varphi \leq 0$, where φ is a formula; an axiom is a condition required to hold for all values of its free variables. Implications between conditions are

expressed by the following device: if φ takes only nonnegative values and $|\psi| \le r_\psi$, then the condition

$$\psi \le \sup_{n\ge 1} \min(r_\psi, n\varphi) \tag{2}$$

says that $\varphi = 0$ implies $\psi \le 0$, since the right side is 0 when $\varphi = 0$ and r_ψ otherwise. The family $\min(r_\psi, n\varphi)$, for $n \ge 1$, is uniformly bounded, which is all that a countable supremum requires.

5.2. **The vocabulary.** The vocabulary $\mathcal{L}$ has three sorts. The two vector sorts are intended to be the closed unit balls of Hilbert spaces; this keeps every predicate bounded.

- ♠ The *Hilbert sort* $\mathbf{H}$ has, for each pair λ, λ' in the field $\mathbb{Q}(i)$ of complex numbers with rational real and imaginary parts, with $|\lambda| + |\lambda'| \le 1$, a binary function $(x, y) \mapsto \lambda x + \lambda' y$; the predicates $\mathrm{Re}\langle x \mid y\rangle$, $\mathrm{Im}\langle x \mid y\rangle$ and $\|x\|$, with values in $[-1, 1]$; and constants e_n, for $n \in \mathbb{N}$. We write 0 for the term $0x + 0x$, and $\langle x \mid y\rangle$ for the pair of predicates $\mathrm{Re}\langle x \mid y\rangle$ and $\mathrm{Im}\langle x \mid y\rangle$.
- ♠ The *sequence sort* $\mathbf{K}$ has the same functions and predicates as $\mathbf{H}$, and one constant ε.
- ♠ The *Solèr sort* $\mathbf{L}$ has the constants 0 and 1, the binary functions $\wedge$ and $\vee$, the unary function $^\perp$, and a predicate $d(a, b)$ with values in $[0, 1]$, the *discrete predicate*.
- ♠ Function symbols connecting the sorts:

$$\pi\colon \mathbf{L} \times \mathbf{H} \to \mathbf{H}, \quad \mathrm{at}\colon \mathbf{H} \to \mathbf{L}, \quad \mathrm{rep}\colon \mathbf{L} \to \mathbf{H},$$
$$\iota_n\colon \mathbf{H} \to \mathbf{K}, \quad \kappa_n\colon \mathbf{K} \to \mathbf{H}\ (n \in \mathbb{N}), \quad \mathrm{span}\colon \mathbf{K} \to \mathbf{L}.$$

- ♠ A predicate μ on $\mathbf{L}$, the *measure*, with values in $[-1, 2]$; the interval is wider than $[0, 1]$ so that axiom (M1), not the vocabulary, confines μ to $[0, 1]$.

For vector terms t_1, t_2 of the same sort, the equation $t_1 \doteq t_2$ abbreviates the condition $\|\frac{1}{2}t_1 - \frac{1}{2}t_2\| = 0$; for terms of the Solèr sort, $t_1 = t_2$ abbreviates $d(t_1, t_2) = 0$. A vector x *belongs* to an element a of the Solèr sort, or is a *member* of a, when $\pi(a, x) \doteq x$. The intended meanings are as follows: $\pi(a, \cdot)$ is the orthogonal projection onto the closed subspace spanned by the members of a; for $x \ne 0$, $\mathrm{at}(x)$ is the element whose members are the multiples of x in the unit ball; $\mathrm{rep}(a)$ is a unit vector belonging to a when $a \ne 0$; (e_n) is an orthonormal basis; ι_n places a vector in position n of a sequence and κ_n reads position n; ε is the sequence $(2^{-n-1}e_n)_n$; and $\mathrm{span}(s)$ is the element whose members are the vectors of the unit ball in the closed span of the entries of s.

Conditions on the values of the measure can be as complicated as Borel sets. For a Borel set $B \subseteq \mathbb{R}$, we write $\mathbf{1}_B$ for its indicator function, which is 1 on B and 0 off B.

Lemma 5.1. *For every Borel set $B \subseteq \mathbb{R}$, there is a formula $\mathbf{1}_B(\mu(a))$ of real-valued $L_{\omega_1,\omega}$, with the single free variable a of the Solèr sort, whose value at every element a of every structure is $\mathbf{1}_B(\mu(a))$.*

Proof. Let $\mathcal{B}$ be the class of sets $B \subseteq \mathbb{R}$ for which such a formula exists. For rational numbers $\alpha \le \beta$, the distance from a real number t to the interval $[\alpha, \beta]$ is $\max(0, \alpha - t, t - \beta)$, a continuous function of t; hence

$$\inf_{n\ge 1} \max\bigl(0,\ 1 - n\max(0, \alpha - \mu(a), \mu(a) - \beta)\bigr)$$

is a formula, and its value is 1 when $\mu(a) \in [\alpha, \beta]$ and 0 otherwise. Thus $\mathcal{B}$ contains the closed intervals with rational endpoints. If $B \in \mathcal{B}$, then $1 - \mathbf{1}_B(\mu(a))$ is a formula for $\mathbb{R} \setminus B$; if $B_n \in \mathcal{B}$ for every $n \in \mathbb{N}$, then the family of formulas $\mathbf{1}_{B_n}(\mu(a))$ is bounded by 1, and

its supremum is a formula for $\bigcup_n B_n$. Hence $\mathcal{B}$ is a σ-algebra. Every open subset of $\mathbb{R}$ is a countable union of closed intervals with rational endpoints, so $\mathcal{B}$ contains every Borel set. ☺

The nesting of countable suprema in the formula $\mathbf{1}_B(\mu(a))$ follows the construction of B from intervals, and may have any countable depth.

Definition 5.2. A structure $\mathcal{M}$ for $\mathcal{L}$ is *standard* when there are complex Hilbert spaces H and K such that the Hilbert sort and the sequence sort of $\mathcal{M}$ are the closed unit balls of H and K, and the functions $\lambda x + \lambda' y$ and the predicates $\mathrm{Re}\langle x \mid y\rangle$, $\mathrm{Im}\langle x \mid y\rangle$ and $\|x\|$ are the restrictions of the operations, the inner product and the norm of H and of K.

Standardness is a restriction on the structures we consider, not a condition imposed by an axiom. Since $\mathcal{L}$ is countable and the theories T_0, T_1 and T of Section 6 below are countable, the downward Löwenheim–Skolem theorem gives them models whose vector sorts are countable, hence not complete; Theorem C applies to such models, whereas the conclusions of Theorem A(1) and Theorem B fail in them (see Remark 10.2).

6. The axioms

The axioms come in groups, one for each sort, one for each family of symbols connecting the sorts, and one for the measure. Together they say that the elements of the Solèr sort name orthogonal projections of the Hilbert sort, ordered and combined as closed subspaces are; that atoms correspond to unit vectors up to a phase; that every sequence of vectors names its closed span; and that μ is σ-additive. Free variables are universally quantified: x, y, z range over $\mathbf{H}$, s, t over $\mathbf{K}$, and a, b, c over $\mathbf{L}$. The letters λ, λ' (ranging over $\mathbb{Q}(i)$, with $|\lambda| + |\lambda'| \leq 1$) and n, m, k, N (ranging over $\mathbb{N}$) are not variables of the logic: an axiom containing them stands for countably many axioms, one for each value. Readings in words follow the axioms where useful.

(H) The Hilbert sort.

(H1) *Inner product space.* For any two terms built from variables by the functions $\lambda x+\lambda' y$ that denote the same $\mathbb{Q}(i)$-linear combination of the variables, the equation between them; the conditions $\langle \lambda x + \lambda' y \mid z\rangle = \lambda\langle x \mid z\rangle + \lambda'\langle y \mid z\rangle$ and $\langle y \mid x\rangle = \overline{\langle x \mid y\rangle}$, each written as two real conditions; and $\|x\| = \sqrt{\max(0, \mathrm{Re}\langle x \mid x\rangle)}$.

(H2) *Separability.* $\langle e_n \mid e_m\rangle = \delta_{nm}$, where δ_{nm} is the Kronecker delta, and

$$\|x\|^2 = \sup_N \min\Big(1, \sum_{n<N} |\langle x \mid e_n\rangle|^2\Big).$$

The constants e_n are orthonormal and satisfy Parseval's identity.

(OL) The Solèr sort.

(OL1) The equations of ortholattices:

$$a \wedge b = b \wedge a, \quad a \wedge (b \wedge c) = (a \wedge b) \wedge c, \quad a \wedge (a \vee b) = a,$$
$$a \vee b = b \vee a, \quad a \vee (b \vee c) = (a \vee b) \vee c, \quad a \vee (a \wedge b) = a,$$
$$a \wedge 0 = 0, \quad a \vee 1 = 1, \quad a^{\perp\perp} = a, \quad (a \wedge b)^{\perp} = a^{\perp} \vee b^{\perp}, \quad a \wedge a^{\perp} = 0.$$

(OL2) *The discrete predicate is equality.* $\min(d(a,b), 1-d(a,b)) = 0$; $d(a,a) = 0$; $|d(a,c)-d(b,c)| \leq d(a,b)$; and $d(a \wedge c, b \wedge c) \leq d(a,b)$, $d(a \vee c, b \vee c) \leq d(a,b)$, $d(a^{\perp}, b^{\perp}) \leq$

$d(a,b)$, $\|\frac{1}{2}\pi(a,x) - \frac{1}{2}\pi(b,x)\| \le d(a,b)$, $\|\frac{1}{2}\mathrm{rep}(a) - \frac{1}{2}\mathrm{rep}(b)\| \le d(a,b)$ and $|\mu(a) - \mu(b)| \le 3\,d(a,b)$.

(P) Each element of the Solèr sort determines a projection.

(P1) $\pi(a, \lambda x + \lambda' y) \doteq \lambda\pi(a,x) + \lambda'\pi(a,y)$.
(P2) $\pi(a, \pi(a,x)) \doteq \pi(a,x)$.
(P3) $\langle \pi(a,x) \mid y\rangle = \langle x \mid \pi(a,y)\rangle$.
(P4) *Extensionality.* $d(a,b) \le \sup_{n\ge 1} \min\Big(1,\, n \sup_x \big\|\frac{1}{2}\pi(a,x) - \frac{1}{2}\pi(b,x)\big\|\Big)$.
Two elements with the same projection are equal.

(Sub) The order and the operations of the Solèr sort are those of subspaces.

(Sub0) $\pi(0,x) \doteq 0$ and $\pi(1,x) \doteq x$.
Only the zero vector belongs to 0*; every vector belongs to* 1.
(Sub1) $\|\frac{1}{2}\pi(b,\pi(a,x)) - \frac{1}{2}\pi(a,x)\| \le d(a \wedge b, a)$, and $d(a \wedge b, a) \le \sup_{n\ge 1} \min\Big(1,\, n \sup_x \big\|\frac{1}{2}\pi(b,\pi(a,x)) - \frac{1}{2}\pi(a,x)\big\|\Big)$.
The relation $a \le b$ holds if and only if every member of a is a member of b.
(Sub2) $\inf_N \sup_{n\ge N} \big\|\frac{1}{2}\pi(a\wedge b, x) - \frac{1}{2}(P_a P_b)^n x\big\| = 0$, where $(P_aP_b)^n x$ is the term obtained by applying $\pi(b,\cdot)$ and then $\pi(a,\cdot)$, n times.
The members of $a \wedge b$ are the common members of a and b.
(Sub3) $\inf_N \sup_{n\ge N} \big\|\frac{1}{4}\pi(a\vee b, x) - \frac{1}{4}x + \frac{1}{4}(P_{a^\perp}P_{b^\perp})^n x\big\| = 0$, the vector inside the norm being denoted by a term built from the functions $\lambda x + \lambda' y$.
The closed subspace spanned by the members of $a \vee b$ is the closure of the sum of the subspaces spanned by the members of a and of b.
(Sub4) $\frac{1}{2}\pi(a^\perp, x) \doteq \frac{1}{2}x - \frac{1}{2}\pi(a,x)$.
The members of $a^\perp$ are the vectors orthogonal to all members of a.

Let $\chi(a)$ be the formula $\sup_b \min\big(d(b,0),\, d(b,a),\, 1 - d(a\wedge b, b)\big)$; it takes the value 1 if some b lies strictly between 0 and a, and 0 otherwise.

(At) Atoms and unit vectors.

(At1) $\pi(\mathrm{at}(x), x) \doteq x$, and $\big|\langle \pi(\mathrm{at}(x),y) \mid \pi(\mathrm{at}(x),z)\rangle\big|^2 = \|\pi(\mathrm{at}(x),y)\|^2\,\|\pi(\mathrm{at}(x),z)\|^2$.
Every vector x belongs to $\mathrm{at}(x)$*, and any two members of* $\mathrm{at}(x)$ *are linearly dependent.*
(At2) $\pi(a, \mathrm{rep}(a)) \doteq \mathrm{rep}(a)$, and $\big|\,\|\mathrm{rep}(a)\| - 1\,\big| \le 1 - d(a,0)$.
The vector $\mathrm{rep}(a)$ *belongs to a, and it is a unit vector when $a \ne 0$.*
(At3) $d(\mathrm{at}(\mathrm{rep}(a)), a) \le \max\big(\chi(a),\, 1 - d(a,0)\big)$.
Every atom a equals $\mathrm{at}(\mathrm{rep}(a))$.
(At4) $\|x\|\,\|y\| - |\langle x \mid y\rangle| \le d(\mathrm{at}(x), \mathrm{at}(y))$, and

$$d(\mathrm{at}(x),\mathrm{at}(y)) \le \sup_{n\ge 1} \min\Big(1,\, n\big(\|x\|\,\|y\| - |\langle x \mid y\rangle| + \big|1 - \|x\|\big| + \big|1 - \|y\|\big|\big)\Big).$$

Two unit vectors determine the same atom if and only if one is a multiple of the other by a complex number of modulus one.

(At5) $\mathrm{at}(0) = 0$, and $d(\mathrm{at}(x), \mathrm{at}(y)) \leq \sup_{n\geq 1} \min\big(1,\, n\|\tfrac{1}{2}x - \tfrac{1}{2}y\|\big)$.

(K) The sequence sort.

(K1) The axioms (H1), written for **K**.

(K2) $\iota_n(\lambda x + \lambda' y) \doteq \lambda\iota_n x + \lambda'\iota_n y$ and $\kappa_n(\lambda s + \lambda' t) \doteq \lambda\kappa_n s + \lambda'\kappa_n t$; $\langle \iota_n x \mid \iota_m y\rangle = \delta_{nm}\langle x \mid y\rangle$; and $\langle \kappa_n s \mid x\rangle = \langle s \mid \iota_n x\rangle$.

(K3) $\|s\|^2 = \sup_N \min\Big(1, \sum_{n<N} \|\kappa_n s\|^2\Big)$.

A sequence is determined by its entries, and its norm is the ℓ^2 norm of its entries.

(K4) $\kappa_n\varepsilon \doteq 2^{-n-1}e_n$.

(Sp) Spans.

(Sp1) $\pi(\mathrm{span}(s), \kappa_n s) \doteq \kappa_n s$.

Every entry of s belongs to $\mathrm{span}(s)$.

(Sp2) $\|\pi(\mathrm{span}(s), x)\|^2 \leq \sup_{m\geq 1} \min\Big(1,\, m \sup_N \min\Big(1, \sum_{n<N} |\langle x \mid \kappa_n s\rangle|^2\Big)\Big)$.

A vector orthogonal to every entry of s is orthogonal to every member of $\mathrm{span}(s)$.

(Sp3) $d(\mathrm{span}(s), \mathrm{span}(t)) \leq \sup_{n\geq 1} \min\big(1,\, n\|\tfrac{1}{2}s - \tfrac{1}{2}t\|\big)$.

Let $\theta(s)$ be the formula $\sup_N \min\big(1, \sum_{n,m<N,\, n\neq m} |\langle \kappa_n s \mid \kappa_m s\rangle|\big)$, which takes values in $[0,1]$ and vanishes exactly when the entries of s are mutually orthogonal.

(M) The measure.

(M1) $0 \leq \mu(a) \leq 1$.

(M2) $\mu(1) = 1$.

(M3) *σ-additivity.*

$$\Big|\, \mu(\mathrm{span}(s)) - \sup_N \min\Big(2, \sum_{n<N} \mu(\mathrm{at}(\kappa_n s))\Big)\Big| \leq \sup_{k\geq 1} \min\big(2,\, k\,\theta(s)\big).$$

If the entries of s are mutually orthogonal, the value of μ at $\mathrm{span}(s)$ *is the sum of its values at the elements* $\mathrm{at}(\kappa_n s)$.

Definition 6.1. The theory T_0 consists of the axioms (H), (OL), (P), (Sub) and (At), without the clause on μ in (OL2), in the vocabulary $\mathcal{L}_0$ obtained from $\mathcal{L}$ by removing the sequence sort with its constant ε, the functions ι_n, κ_n and span, and the predicate μ. The theory T_1 consists of T_0 together with (K) and (Sp), in the vocabulary $\mathcal{L}$ without μ; the theory T consists of T_1 together with (M) and the clause on μ in (OL2).

Lemma 6.2. *In a model of T_0, two elements a, b of the Solèr sort are equal if and only if $d(a,b) = 0$, and two vectors x, y are equal if and only if $\|\tfrac{1}{2}x - \tfrac{1}{2}y\| = 0$; the same holds in the sequence sort of a model of T_1.*

Proof. By (H1), (K1), (OL2), (P1)–(P3), (At5), (K2) and (Sp3), and by the Cauchy–Schwarz inequality for the predicates, every atomic formula takes the same values at a and b when $d(a,b)=0$, and at x and y when $\|\frac{1}{2}x-\frac{1}{2}y\|=0$; since models are reduced, the elements in question are equal. ☺

Lemma 6.3. *Let $\mathcal{M}$ be a model of T_0. There is a complex Hilbert space $\bar{E}$ such that the Hilbert sort of $\mathcal{M}$ can be identified with a subset B of the closed unit ball of $\bar{E}$ whose $\mathbb{Q}(i)$-linear span E is dense, the functions $\lambda x+\lambda' y$ and the predicates being the restrictions of those of $\bar{E}$. For every element a of the Solèr sort, $\pi(a,\cdot)$ extends uniquely to an orthogonal projection $\bar{P}_a$ of $\bar{E}$. If $\mathcal{M}$ is standard, then B is the whole closed unit ball of $\bar{E}$.*

Proof. On pairs (k,x), with $k\geq 1$ an integer and $x\in B$, read as kx, put $(k,x)\sim(l,y)$ when $\|\frac{1}{2l}x-\frac{1}{2k}y\|=0$; define the sum by $(k,x)+(l,y)=(2kl,\frac{1}{2l}x+\frac{1}{2k}y)$ and the form by $\langle(k,x)\mid(l,y)\rangle=kl\langle x\mid y\rangle$; multiplication by elements of $\mathbb{Q}(i)$ is defined similarly. Axiom (H1) makes the quotient a $\mathbb{Q}(i)$-vector space E with a positive definite hermitian form, and Lemma 6.2 makes the map from B to E injective; we let $\bar{E}$ be the completion of E. By (P1)–(P3), the map $\pi(a,\cdot)$ extends to a $\mathbb{Q}(i)$-linear idempotent map Q_a on E, symmetric with respect to the form; hence $\|Q_a x\|^2=\langle x\mid Q_a x\rangle\leq\|x\|\|Q_a x\|$, the map Q_a is contractive, and it extends by continuity to a self-adjoint idempotent $\bar{P}_a$ of $\bar{E}$. In a standard model, the set B is the closed unit ball of a Hilbert space, which is complete and spanned by B; that space is therefore $\bar{E}$. ☺

7. Standard models of T_0: pre-Solèr lattices

In a standard model of T_0, the theory without the sequence sort and without the measure, the Solèr sort is a pre-Solèr lattice whose image is dense, in the order-theoretic sense of Section 1, in the lattice of closed subspaces of the Hilbert space; and the σ-additive states on it, defined below, already come from density operators. Throughout this section, $\mathcal{M}$ is a standard model of T_0. We write H for the Hilbert space whose closed unit ball is the Hilbert sort of $\mathcal{M}$ (see Definition 5.2), $\mathbf{L}^{\mathcal{M}}$ for its Solèr sort, P_a for the projection $\bar{P}_a$ of Lemma 6.3, and S_a for the range of P_a, the closed subspace spanned by the members of a.

Proposition 7.1. *The space H is a separable infinite-dimensional complex Hilbert space, and $(e_n)_{n\in\mathbb{N}}$ is an orthonormal basis of H.*

Proof. By (H2) the vectors e_n are orthonormal, and Parseval's identity shows that every vector of the unit ball, and hence of H, lies in their closed span; thus no nonzero vector is orthogonal to all of them. ☺

Theorem 7.2 (von Neumann's alternating projection theorem [vN49, Lemma 22]**).** *Let U and V be closed subspaces of a Hilbert space, and let P_U, P_V and $P_{U\cap V}$ be the orthogonal projections onto U, V and $U\cap V$. Then $(P_U P_V)^n x$ converges in norm to $P_{U\cap V}x$ for every vector x.*

Proposition 7.3. *The map $\tau\colon a\mapsto S_a$ is an embedding of ortholattices from $\mathbf{L}^{\mathcal{M}}$ into $L(H)$; that is, it is injective and satisfies*

$$S_0=\{0\},\quad S_1=H,\quad S_{a\wedge b}=S_a\cap S_b,\quad S_{a\vee b}=\overline{S_a+S_b},\quad S_{a^\perp}=S_a^\perp.$$

Moreover, the relation $a\leq b$ holds if and only if $S_a\subseteq S_b$.

Proof. Axiom (Sub0) gives the first two equalities, and axiom (Sub4) gives $P_{a^\perp}=\mathrm{Id}-P_a$, whose range is $S_a^\perp$. Combined with Theorem 7.2, axiom (Sub2) shows that $P_{a\wedge b}=P_{S_a\cap S_b}$, and axiom (Sub3) that $P_{a\vee b}=\mathrm{Id}-P_{S_a^\perp\cap S_b^\perp}$, whose range is $(S_a^\perp\cap S_b^\perp)^\perp=\overline{S_a+S_b}$. If $S_a=S_b$,

then $P_a = P_b$, and axiom (P4) gives $a = b$. Finally, axiom (Sub1) shows that $a \wedge b = a$ exactly when $P_b P_a = P_a$, that is, when $S_a \subseteq S_b$. ☺

Proposition 7.4. *For every nonzero x in the unit ball, the subspace $S_{\mathrm{at}(x)}$ is $\mathbb{C}x$. The atoms of $\mathbf{L}^{\mathcal{M}}$ are exactly the elements $\mathrm{at}(x)$ with $x \neq 0$, and $x \mapsto \mathrm{at}(x)$ induces a bijection from the unit sphere of H, modulo multiplication by complex numbers of modulus one, onto the set of atoms of $\mathbf{L}^{\mathcal{M}}$, whose inverse sends an atom a to the class of $\mathrm{rep}(a)$.*

Proof. By (At1), the vector x lies in $S_{\mathrm{at}(x)}$, and any two vectors of $S_{\mathrm{at}(x)}$ attain equality in the Cauchy–Schwarz inequality, hence are linearly dependent; therefore $S_{\mathrm{at}(x)} = \mathbb{C}x$. Since this ray is an atom of $L(H)$ and τ is an order embedding, $\mathrm{at}(x)$ is an atom of $\mathbf{L}^{\mathcal{M}}$. If a is an atom of $\mathbf{L}^{\mathcal{M}}$, then axiom (At3) gives $a = \mathrm{at}(\mathrm{rep}(a))$. For unit vectors x and y, axiom (At4) gives $\mathrm{at}(x) = \mathrm{at}(y)$ if and only if $|\langle x \mid y\rangle| = 1$, that is, if and only if y is a multiple of x by a complex number of modulus one. ☺

Theorem 7.5. *Let $\mathcal{M}$ be a standard model of T_0. Its Solèr sort $\mathbf{L}^{\mathcal{M}}$ is a pre-Solèr lattice, and the image of the map $\tau\colon a \mapsto S_a$ is join-dense and meet-dense in $L(H)$, which is therefore isomorphic to the MacNeille completion of $\mathbf{L}^{\mathcal{M}}$. The image of τ contains every finite-dimensional subspace of H and the orthogonal complement of each, and τ maps every join that exists in $\mathbf{L}^{\mathcal{M}}$ to the corresponding join in $L(H)$.*

Proof. If F is a finite-dimensional subspace with a basis $x_1, \ldots, x_k$ of unit vectors, then $\tau(\mathrm{at}(x_1) \vee \cdots \vee \mathrm{at}(x_k)) = \overline{F} = F$ by Propositions 7.3 and 7.4, and the orthogonal complement $F^\perp$ is $\tau\big((\mathrm{at}(x_1) \vee \cdots \vee \mathrm{at}(x_k))^\perp\big)$. The lattice $\mathbf{L}^{\mathcal{M}}$ is orthomodular because $\tau(\mathbf{L}^{\mathcal{M}})$ is a subortholattice of the orthomodular lattice $L(H)$. If $a \neq 0$, the atoms below a are the elements $\mathrm{at}(x)$ with $x \in S_a$ and $0 < \|x\| \leq 1$; any upper bound b of them in $\mathbf{L}^{\mathcal{M}}$ satisfies $S_b \supseteq S_a$, hence $b \geq a$, and a is their join; thus $\mathbf{L}^{\mathcal{M}}$ is atomistic. For the covering property, let $\mathrm{at}(x)$ be an atom not below a; then $x \notin S_a$, and Proposition 7.3 and Lemma 2.7 give $\tau(a \vee \mathrm{at}(x)) = S_a + \mathbb{C}x$, in which S_a has codimension one; since τ is an order embedding, no element lies strictly between a and $a \vee \mathrm{at}(x)$. The join of two distinct atoms corresponds to a plane, which contains infinitely many rays, all in $\tau(\mathbf{L}^{\mathcal{M}})$; this gives (I). The atoms $\mathrm{at}(e_n)$ give (D), and condition (S) holds with $p_n = \mathrm{at}(e_n)$ and $c_n = \mathrm{at}(\frac{1}{2}(e_n + e_{n+1}))$, by (1) and the computation following it.

Every closed subspace S of H is the join in $L(H)$ of its rays, and it equals $\bigcap\{y^\perp : y \in S^\perp,\ 0 < \|y\| \leq 1\}$, where $y^\perp = S_{\mathrm{at}(y)^\perp}$ lies in the image of τ; these two facts are join-density and meet-density, and Proposition 1.5 identifies $L(H)$ with the MacNeille completion of $\mathbf{L}^{\mathcal{M}}$. Finally, let $A \subseteq \mathbf{L}^{\mathcal{M}}$ have a join c in $\mathbf{L}^{\mathcal{M}}$, and let W be the closed span of $\bigcup_{a \in A} S_a$; clearly $W \subseteq S_c$. If y is a vector of $S_c \cap W^\perp$ with $0 < \|y\| \leq 1$, then every $a \in A$ lies below $\mathrm{at}(y)^\perp$, hence so does c, and y is orthogonal to itself, which is absurd. Since $S_c = W + (S_c \cap W^\perp)$, it follows that $S_c = W$. ☺

Example 7.6. Let L_{fc} (for "finite dimension or finite codimension") be the set of finite-dimensional subspaces of ℓ^2 and their orthogonal complements. With d the discrete metric, $\pi(S, x) = P_S x$, $\mathrm{at}(x) = \mathbb{C}x$, a unit vector $\mathrm{rep}(S) \in S$ chosen for each $S \neq \{0\}$, $\mathrm{rep}(\{0\}) = 0$, and the standard unit vectors as (e_n), the lattice L_{fc} is the Solèr sort of a standard model of T_0, although it is not complete. The axioms of T_0 constrain only the elements that the Solèr sort contains; none of them demands an element whose members span a given closed subspace of infinite dimension and infinite codimension. The sequence sort, which the axioms of T_1 govern, is introduced to make that demand.

A *density operator* on H is a positive operator ρ whose trace $\operatorname{Tr}\rho = \sum_k \langle \rho u_k \mid u_k\rangle$, computed in any orthonormal basis $(u_k)_{k\in\mathbb{N}}$ of H, equals 1; quantum mechanics represents the states of a system by density operators. A *σ-additive state* on an ortholattice Λ is a map $\nu\colon \Lambda \to [0,1]$ with $\nu(1) = 1$ such that $\nu(\bigvee_n a_n) = \sum_n \nu(a_n)$ for every finite or countable family of mutually orthogonal elements whose join exists in Λ.

Theorem 7.7 (Gleason's theorem for T_0). *Let $\mathcal{M}$ be a standard model of T_0, let H be the Hilbert space whose closed unit ball is its Hilbert sort, and, for each element a of its Solèr sort $\mathbf{L}^{\mathcal{M}}$, let P_a be the orthogonal projection onto the closed subspace spanned by the members of a. The σ-additive states on $\mathbf{L}^{\mathcal{M}}$ are exactly the maps $a \mapsto \operatorname{Tr}(\rho P_a)$, where ρ ranges over the density operators on H, and each such state determines its density operator.*

Proof. If the elements a_n, for n in a finite or countable set, are mutually orthogonal with join c in $\mathbf{L}^{\mathcal{M}}$, then by Theorem 7.5 the subspace S_c is the closure of $\sum_n S_{a_n}$; hence $P_c x = \sum_n P_{a_n} x$ for every $x \in H$, and $\operatorname{Tr}(\rho P_c) = \sum_n \operatorname{Tr}(\rho P_{a_n})$; moreover $\operatorname{Tr}(\rho P_1) = \operatorname{Tr}\rho = 1$. Conversely, let ν be a σ-additive state; additivity on the pair $1, 0$ gives $\nu(0) = 0$. For a unit vector x, put $f(x) = \nu(\operatorname{at}(x))$. For $a \neq 0$ and an orthonormal basis (u_k) of S_a, which is countable, the atoms $\operatorname{at}(u_k)$ are mutually orthogonal and their join in $\mathbf{L}^{\mathcal{M}}$ is a, because every upper bound b of them satisfies $S_b \supseteq S_a$; hence $\nu(a) = \sum_k f(u_k)$. Taking $a = 1$ shows that f is a *frame function* of weight 1: its sum over every orthonormal basis of H is 1. Since H is separable of dimension at least three, Gleason's theorem [Gle57] gives a unique density operator ρ with $f(x) = \langle \rho x \mid x\rangle$ for every unit vector x, and then $\nu(a) = \sum_k \langle \rho u_k \mid u_k\rangle = \operatorname{Tr}(\rho P_a)$. ☻

8. Standard models of T_1 and the proofs of Theorems A and B

The axioms of T_0 cannot compel the Solèr sort to contain a given closed subspace of infinite dimension and infinite codimension, as the lattice L_{fc} of Example 7.6 shows. The sequence sort supplies what is missing: every closed subspace of a separable Hilbert space is the closed span of a square-summable sequence, and the function span names it. In this section, $\mathcal{M}$ is a standard model of T_1; we write H and K for the Hilbert spaces whose closed unit balls are its Hilbert sort and its sequence sort, and $\mathbf{L}^{\mathcal{M}}$ for its Solèr sort. Since T_1 contains T_0, the results of Section 7 apply to $\mathcal{M}$, and P_a, S_a and τ are as there.

Lemma 8.1. *The maps κ_n extend to bounded linear maps from K to H, and $s \mapsto (\kappa_n s)_{n\in\mathbb{N}}$ is a linear isometry of K onto the space $\ell^2(\mathbb{N}, H)$ of sequences $(x_n)_{n\in\mathbb{N}}$ in H with $\sum_n \|x_n\|^2 < \infty$.*

Proof. By (K2) and (K3), the map is linear and isometric, and $\kappa_m \iota_n x = \delta_{mn} x$. Given $(x_n) \in \ell^2(\mathbb{N}, H)$, the partial sums $\sum_{n<N} \iota_n x_n$ form a Cauchy sequence in K, by (K2); since K is complete, they converge to some s, and $\kappa_n s = x_n$ by the continuity of κ_n. ☻

Lemma 8.2. *For every s in the unit ball of K, the subspace $S_{\operatorname{span}(s)}$ is the closed span of the entries $\kappa_n s$.*

Proof. Let W be that closed span. By (Sp1), the subspace $S_{\operatorname{span}(s)}$ contains every entry of s; hence $W \subseteq S_{\operatorname{span}(s)}$. If $x \perp W$, axiom (Sp2) gives $P_{\operatorname{span}(s)} x = 0$; hence $S_{\operatorname{span}(s)} \subseteq W^{\perp\perp} = W$. ☻

Theorem 8.3. *Let $\mathcal{M}$ be a standard model of T_1, and let H be the Hilbert space whose closed unit ball is its Hilbert sort. Then H is a separable infinite-dimensional complex Hilbert space, and the map $\tau\colon a \mapsto S_a$ is an isomorphism of ortholattices from the Solèr sort of $\mathcal{M}$ onto $L(H)$; in particular, that Solèr sort is a Solèr lattice. Conversely, for every separable infinite-dimensional complex Hilbert space H', there is a standard model of T_1 whose Hilbert sort is the closed unit ball of H' and whose Solèr sort is $L(H')$.*

Proof. By Proposition 7.1, the space H is separable and infinite-dimensional. By Proposition 7.3, the map τ is an embedding; it remains to be shown that it is onto. Let S be a closed subspace of H, and let $(u_n)_{n\in\mathbb{N}}$ enumerate an orthonormal basis of S, which is countable by Proposition 7.1, followed by zeros if the basis is finite. By Lemma 8.1 there is s in the unit ball of K with $\kappa_n s = 2^{-n-1}u_n$ for all n, since $\sum_n 4^{-n-1} < 1$, and Lemma 8.2 gives $S_{\mathrm{span}(s)} = S$.

For the converse, interpret the Solèr sort as $L(H')$, with d the discrete metric ($d(S,S') = 1$ when $S \neq S'$), $\pi(S,x) = P_S x$ and $\mathrm{at}(x) = \mathbb{C}x$; let $\mathrm{rep}(S)$ be a unit vector chosen in each $S \neq \{0\}$, and let $\mathrm{rep}(\{0\})$ be 0; let (e_n) be an orthonormal basis of H'; interpret the sequence sort as the unit ball of $\ell^2(\mathbb{N}, H')$, with $\iota_n x$ the sequence with x in position n and 0 elsewhere, $\kappa_n s$ the entry of s in position n, and $\varepsilon = (2^{-n-1}e_n)_n$; and let $\mathrm{span}(s)$ be the closed span of the entries of s. Every axiom holds by construction, axioms (Sub2) and (Sub3) by Theorem 7.2. ☺

Proof of Theorem A. Let $\mathcal{M}$ be a standard model of T; it is in particular a standard model of T_1, and by Theorem 8.3 the space H is a separable infinite-dimensional complex Hilbert space and τ is an isomorphism of $\mathbf{L}^{\mathcal{M}}$ onto $L(H)$.

Step 1: $\mu(0) = 0$. Let s be the zero element of the sequence sort. Its entries are 0; hence $\theta(s) = 0$, and Lemma 8.2 gives $S_{\mathrm{span}(s)} = \{0\}$, whence $\mathrm{span}(s) = 0$ by the injectivity of τ. Moreover, axiom (At5) gives $\mathrm{at}(\kappa_n s) = \mathrm{at}(0) = 0$. Axiom (M3) gives $\mu(0) = \sup_N \min(2, N\mu(0))$; if $\mu(0)$ were positive, the right side would be 2, contradicting (M1).

Step 2: sums along orthonormal bases. Let a be an element of $\mathbf{L}^{\mathcal{M}}$, and let $(u_k)_{k\in\mathbb{N}}$ enumerate an orthonormal basis of S_a, followed by zeros if it is finite. By Lemma 8.1 there is s in the unit ball of K with $\kappa_k s = 2^{-k-1}u_k$. Its entries are mutually orthogonal, whence $\theta(s) = 0$; by Lemma 8.2 and the injectivity of τ, the element $\mathrm{span}(s)$ is a. For $u_k \neq 0$, Proposition 7.4 gives $S_{\mathrm{at}(2^{-k-1}u_k)} = \mathbb{C}u_k$, hence $\mathrm{at}(\kappa_k s) = \mathrm{at}(u_k)$; for $u_k = 0$, the element $\mathrm{at}(\kappa_k s)$ is 0, whose measure is 0 by Step 1. Axiom (M3) and $\mu(a) \leq 1$ give

$$\mu(a) = \sum_k \mu(\mathrm{at}(u_k)). \tag{$*$}$$

Step 3: application of Gleason's theorem. For a unit vector x, put $f(x) = \mu(\mathrm{at}(x))$. Applying $(*)$ to $a = 1$ and using (M2), we obtain $\sum_k f(u_k) = 1$ for every orthonormal basis (u_k) of H. Arguing as in the last three sentences of the proof of Theorem 7.7, with μ in place of ν and $(*)$ in place of the decomposition used there, we obtain a unique density operator ρ on H with $\mu(a) = \mathrm{Tr}(\rho P_a)$ for every $a \in \mathbf{L}^{\mathcal{M}}$.

The converse. Given a separable infinite-dimensional complex Hilbert space H' and a density operator ρ on H', expand the model of Theorem 8.3 for H' by $\mu(a) = \mathrm{Tr}(\rho P_a)$. The clause on μ in (OL2) holds because d is the discrete metric and $|\mu(a) - \mu(b)| \leq 1$. Axioms (M1) and (M2) hold because $\mathrm{Tr}(\rho P) \in [0,1]$ for every orthogonal projection P and $\mathrm{Tr}\,\rho = 1$. For (M3), suppose that the entries $x_n = \kappa_n s$ are mutually orthogonal. Then

$\mathrm{span}(s)$ is their closed span, and, with $\hat{x}_n = x_n/\|x_n\|$, we have $\mathrm{Tr}(\rho P_{\mathrm{span}(s)}) = \sum_{x_n \neq 0} \langle \rho \hat{x}_n \mid \hat{x}_n \rangle = \sum_n \mu(\mathrm{at}(x_n)) \leq 1$; the left side of (M3) is therefore 0. If the entries are not mutually orthogonal, the right side of (M3) is 2, while the left side is at most 2. ☺

The proof of Theorem B rests on the fact that a density operator, unlike a two-valued measure, takes every value in $[0,1]$ on closed subspaces.

Lemma 8.4. *Let H be a separable infinite-dimensional complex Hilbert space and ρ a density operator on H. Then $\{\mathrm{Tr}(\rho P_S) : S \in L(H)\} = [0,1]$.*

Proof. The values 0 and 1 are attained at $\{0\}$ and H. Fix α with $0 < \alpha < 1$. Since ρ is a positive compact operator of trace 1, there are orthonormal eigenvectors $v_1, v_2, \ldots$ (finitely or infinitely many) whose eigenvalues $\eta_1 \geq \eta_2 \geq \cdots > 0$ are the nonzero eigenvalues of ρ, repeated according to multiplicity, with $\sum_i \eta_i = 1$. Write $\eta_{\leq k} = \eta_1 + \cdots + \eta_k$, with $\eta_{\leq 0} = 0$, and choose k with $\eta_{\leq k} < \alpha \leq \eta_{\leq k+1}$. On the unit vectors of $W = \{v_1, \ldots, v_{k+1}\}^\perp$, the quantity $\langle \rho x \mid x \rangle$ has infimum 0: either W meets the kernel of ρ in a nonzero vector, or the eigenvalues η_i with $i > k+1$ tend to 0. Choose a unit vector $u \in W$ with $\langle \rho u \mid u \rangle < \alpha - \eta_{\leq k}$. For $0 \leq \vartheta \leq 1$, let w_ϑ be the unit vector $\sqrt{1-\vartheta}\, v_{k+1} + \sqrt{\vartheta}\, u$, and let S_ϑ be the subspace spanned by $v_1, \ldots, v_k$ and w_ϑ. Since $\rho v_{k+1} = \eta_{k+1} v_{k+1}$ and u is orthogonal to $v_1, \ldots, v_{k+1}$, the cross terms vanish, and

$$\mathrm{Tr}(\rho P_{S_\vartheta}) = \eta_{\leq k} + (1-\vartheta)\eta_{k+1} + \vartheta \langle \rho u \mid u \rangle,$$

a continuous function of ϑ that equals $\eta_{\leq k+1} \geq \alpha$ at $\vartheta = 0$ and is smaller than α at $\vartheta = 1$. By the intermediate value theorem, it takes the value α. ☺

Proof of Theorem B. In a standard model of T, Theorem A gives $\mu(a) = \mathrm{Tr}(\rho P_a)$ for some density operator ρ on H, and $a \mapsto S_a$ is onto $L(H)$; by Lemma 8.4, the range of μ is $[0,1]$. Since B meets $[0,1]$, some element a satisfies $\mu(a) \in B$, and the value of $\mathbf{1}_B(\mu(a))$ at that element is 1; hence the axiom of Theorem B fails. ☺

9. The proof of Theorem C

The proof of Theorem C, which says that no model of T carries a measure with only the values 0 and 1, works in an arbitrary model of T, standard or not, and uses, besides the basis (e_n), only the eighteen vectors of Table 1. Throughout this section, $\mathcal{M}$ is a model of T in which μ takes only the values 0 and 1; we derive a contradiction, which proves Theorem C. Let $\bar{E}$ and $\bar{P}_a$ be as in Lemma 6.3. By (H2), the vectors e_n form an orthonormal basis of $\bar{E}$: Parseval's identity puts every vector of B, and hence of $\bar{E}$, in their closed span. The proof rests on three facts that hold in every model of T, standard or not.

Lemma 9.1. *(1) For every nonzero x in the Hilbert sort and every $\lambda \in \mathbb{Q}(i)$ with $0 < |\lambda| \leq 1$, we have $\mathrm{at}(\lambda x) = \mathrm{at}(x)$, and $\bar{P}_{\mathrm{at}(x)}$ is the projection onto $\mathbb{C}x$.*

(2) Let s be an element of the sequence sort, and let F be the closed span of its entries in $\bar{E}$. If every e_n lies in F or in $F^\perp$, then $\bar{P}_{\mathrm{span}(s)}$ is the projection onto F.

(3) The measure satisfies $\mu(0) = 0$, and $\mathrm{span}(\varepsilon) = 1$.

Proof. (1) By (At1) and continuity, the range of $\bar{P}_{\mathrm{at}(x)}$ contains x, and any two of its vectors are linearly dependent; hence it is $\mathbb{C}x$, and likewise for λx. The two projections coincide, and axiom (P4) gives $\mathrm{at}(\lambda x) = \mathrm{at}(x)$.

(2) By (Sp1) and linearity, the projection $\bar{P}_{\mathrm{span}(s)}$ fixes every entry of s; hence its range contains F. If $e_n \in F^\perp$, then the vector e_n is orthogonal to every entry, and axiom (Sp2) gives $\bar{P}_{\mathrm{span}(s)} e_n = 0$. The vectors e_n that lie in $F^\perp$ span a dense subspace of $F^\perp$, because the

e_n form an orthonormal basis and each lies in F or in $F^\perp$; by continuity, $\bar{P}_{\mathrm{span}(s)}$ vanishes on $F^\perp$, and its range is F.

(3) For $s = 0$, every vector of the Hilbert sort is orthogonal to the entries, and axiom (Sp2) gives $\bar{P}_{\mathrm{span}(0)} = 0 = \bar{P}_0$; axiom (P4) gives $\mathrm{span}(0) = 0$, and the argument of Step 1 in the proof of Theorem A gives $\mu(0) = 0$. For $s = \varepsilon$, the entries are $2^{-n-1}e_n$ by (K4); the space F is then $\bar{E}$, and part (2) gives $\bar{P}_{\mathrm{span}(\varepsilon)} = \mathrm{Id} = \bar{P}_1$, whence $\mathrm{span}(\varepsilon) = 1$ by (P4). ☺

basis	vectors (coordinates with respect to g_0, g_1, g_2, g_3)			
1	$(0,0,0,1)$	$(0,0,1,0)$	$(1,1,0,0)$	$(1,-1,0,0)$
2	$(0,0,0,1)$	$(0,1,0,0)$	$(1,0,1,0)$	$(1,0,-1,0)$
3	$(1,-1,1,-1)$	$(1,-1,-1,1)$	$(1,1,0,0)$	$(0,0,1,1)$
4	$(1,-1,1,-1)$	$(1,1,1,1)$	$(1,0,-1,0)$	$(0,1,0,-1)$
5	$(0,0,1,0)$	$(0,1,0,0)$	$(1,0,0,1)$	$(1,0,0,-1)$
6	$(1,-1,-1,1)$	$(1,1,1,1)$	$(1,0,0,-1)$	$(0,1,-1,0)$
7	$(1,1,-1,1)$	$(1,1,1,-1)$	$(1,-1,0,0)$	$(0,0,1,1)$
8	$(1,1,-1,1)$	$(-1,1,1,1)$	$(1,0,1,0)$	$(0,1,0,-1)$
9	$(1,1,1,-1)$	$(-1,1,1,1)$	$(1,0,0,1)$	$(0,1,-1,0)$

TABLE 1. The configuration of Cabello, Estebaranz and García-Alcaine [CEGA96]: nine orthogonal bases of a four-dimensional space, formed by eighteen vectors, each of which occurs in exactly two bases.

Proof of Theorem C. The entries of ε are mutually orthogonal, whence $\theta(\varepsilon) = 0$, and Lemma 9.1(1) gives $\mathrm{at}(2^{-n-1}e_n) = \mathrm{at}(e_n)$; by (M2), (K4), (M3) and Lemma 9.1(3), we obtain

$$1 = \mu(1) = \mu(\mathrm{span}(\varepsilon)) = \sum_n \mu(\mathrm{at}(e_n)).$$

As μ takes only the values 0 and 1, there is exactly one $n_0 \in \mathbb{N}$ with $\mu(\mathrm{at}(e_{n_0})) = 1$. Let $j \in \mathbb{N}$ satisfy $4j \le n_0 < 4j + 4$, put $g_i = e_{4j+i}$ for $i = 0, 1, 2, 3$, and let F be the span of $g_0, \dots, g_3$.

Let $\mathcal{B}$ be an orthogonal basis of F consisting of vectors $v_0, \dots, v_3$ whose coordinates with respect to $g_0, \dots, g_3$ lie in $\{0, 1, -1\}$. Each vector $\frac{1}{4}v_i$ has coordinates whose absolute values sum to at most 1, and it is therefore denoted by a term built from the constants e_n by the functions $\lambda x + \lambda' y$. Hence the sequence $\sum_{i<4} \iota_i(\frac{1}{16}v_i)$ is an element $s_{\mathcal{B}}$ of the sequence sort, denoted by the term $\frac{1}{2}\big(\frac{1}{2}\iota_0(\frac{1}{4}v_0) + \frac{1}{2}\iota_1(\frac{1}{4}v_1)\big) + \frac{1}{2}\big(\frac{1}{2}\iota_2(\frac{1}{4}v_2) + \frac{1}{2}\iota_3(\frac{1}{4}v_3)\big)$. The entries of $s_{\mathcal{B}}$ are $\frac{1}{16}v_0, \dots, \frac{1}{16}v_3$ and 0, which are mutually orthogonal; hence $\theta(s_{\mathcal{B}}) = 0$. Each e_n lies in F or in $F^\perp$, and Lemma 9.1(2) shows that $\bar{P}_{\mathrm{span}(s_{\mathcal{B}})}$ is the projection onto F. For a vector v with coordinates in $\{0, 1, -1\}$, we write $\mathrm{at}(v)$ for $\mathrm{at}(\frac{1}{16}v)$; by Lemma 9.1(1), this element does not change when the sign of v is changed. By (P4), all the elements $\mathrm{span}(s_{\mathcal{B}})$ are equal to one element a_F of the Solèr sort, and axiom (M3), Lemma 9.1(1) and $\mu(0) = 0$ give

$$\mu(a_F) = \sum_{v \in \mathcal{B}} \mu(\mathrm{at}(v)).$$

For the basis $g_0, \dots, g_3$ this sum contains $\mu(\mathrm{at}(e_{n_0})) = 1$ and three zeros; hence $\mu(a_F) = 1$, and every such basis $\mathcal{B}$ contains exactly one vector v with $\mu(\mathrm{at}(v)) = 1$.

Apply this to the nine bases of Table 1, whose vectors span pairwise distinct rays. Count the pairs $(\mathcal{B}, v)$ in which $\mathcal{B}$ is one of the nine bases, $v \in \mathcal{B}$ and $\mu(\mathrm{at}(v)) = 1$. Each basis contributes exactly one pair, and the count is 9. On the other hand, each of the eighteen vectors occurs in exactly two bases, and each vector v with $\mu(\mathrm{at}(v)) = 1$ therefore contributes exactly two pairs; the count is even. This contradiction proves the theorem. ☻

Combined with axiom (M1), Theorem C gives the following.

Corollary 9.2 (No hidden variables). *In every model of T, standard or not, some element a of the Solèr sort satisfies $0 < \mu(a) < 1$.*

Proof. By (M1), the measure takes values in $[0, 1]$, and by Theorem C it does not take only the values 0 and 1. ☻

10. Remarks

The remarks that follow compare the theory with its predecessor, explain why standardness cannot be dropped from Theorems A and B, and describe the other choices of scalars, the logical form of the axioms and their redundancies.

Remark 10.1 (Solèr triples). The structures considered here refine the Solèr triples of Aubone, Dueñez and Iovino [ADI], in which a Hilbert space and a Solèr lattice are linked by a membership relation. First, in a Solèr triple the lattice is assumed complete, whereas here completeness of the Solèr sort is a consequence of the axioms of T_1 in standard models (see Theorem 8.3). Second, the membership relation is replaced by the projection π: a vector x belongs to a exactly when $\|\frac{1}{2}x - \frac{1}{2}\pi(a, x)\| = 0$. In real-valued logic an existential quantifier becomes an infimum. Read in this way, the condition "$x = y + z$ for some member y of a and some member z of b" holds exactly at the vectors whose distance from $S_a + S_b$ is 0, that is, on the closure of $S_a + S_b$, which is the join of S_a and S_b in $L(H)$. Third, the functions at and rep provide the bijection between the rays of H, which correspond to the unit vectors up to a phase (a complex factor of modulus one), and the atoms of the Solèr sort (see Proposition 7.4).

Remark 10.2 (Nonstandard models). Standardness cannot be dropped from Theorem B or from Theorem A(1): by the downward Löwenheim–Skolem theorem, T has countable models, in which the range of μ is countable and the Solèr sort, being countable, is not of the form $L(H)$. Without the sequence sort, even the conclusion of Gleason's theorem can fail. Let E_0 be the space of finitely supported complex sequences with the inner product of ℓ^2, and let Λ_0 be the lattice of finite-dimensional subspaces of E_0 and their orthogonal complements in E_0 (see Example 4.3). Take the closed unit ball of E_0 as Hilbert sort and Λ_0 as Solèr sort, with the orthogonal projections onto its elements, $\mathrm{at}(x) = \mathbb{C}x$, a choice of rep, and the standard unit vectors as (e_n). The resulting structure is a model of T_0 that is not standard. Every maximal orthonormal family of E_0 is countable and has join 1 in Λ_0; a σ-additive state on Λ_0 would therefore yield a frame function on E_0, a function on the unit vectors whose sum over every maximal orthonormal family is 1 and whose restriction to the unit vectors of every finite-dimensional subspace is a frame function of that subspace. By a theorem of Dvurečenskij [Dvu95, Theorem 3.1] (compare Hamhalter and Pták [HP87]), an inner product space that carries such a function is complete, and E_0 is not. Hence Λ_0 has no σ-additive states, although density operators on ℓ^2 abound, and the conclusion of Theorem 7.7 fails for this nonstandard model.

The axioms of Theorem B exclude, one Borel set at a time, every range other than $[0, 1]$ in standard models; but no sentence says that the range of μ is disconnected. Indeed, suppose

that a sentence σ took the value 0 exactly in the models of T in which the range of μ is disconnected. In a standard model $\mathcal{M}$ of T, the range of μ is $[0,1]$ by Theorem B, and so the value of σ in $\mathcal{M}$ is not 0. By the downward Löwenheim–Skolem theorem, $\mathcal{M}$ has a countable elementary substructure $\mathcal{N}$ for a countable fragment containing T and σ; the value of σ in $\mathcal{N}$ is also not 0, yet the range of μ in $\mathcal{N}$ is countable, and therefore disconnected, a contradiction.

Whether a nonstandard model of T can carry a measure that is not of the form $a \mapsto \mathrm{Tr}(\rho \bar{P}_a)$ we leave open. Axiom (K4) is essential to Theorem C. Take the closed unit ball of ℓ^2 as Hilbert sort, the lattice L_{fc} of Example 7.6, with the same interpretations of d, π, at, rep and (e_n), as Solèr sort, the finitely supported sequences in the closed unit ball of $\ell^2(\mathbb{N}, \ell^2)$ as sequence sort, with ι_n and κ_n as in Theorem 8.3, $\varepsilon = 0$, and $\mathrm{span}(s)$ the span of the entries of s, and let μ vanish on the finite-dimensional subspaces and equal 1 on their orthogonal complements. Every axiom of T except (K4) holds, because the instances of (M3) then concern only finite orthogonal families, and μ takes only the values 0 and 1.

Remark 10.3 (Scalars). The vocabulary fixes the complex numbers as scalars. For real Hilbert spaces, replace $\mathbb{Q}(i)$ by $\mathbb{Q}$ and omit the imaginary parts; for quaternionic spaces, use the quaternions with rational coordinates, acting on the left, and the four real components of the form; in (At4), "complex number of modulus one" becomes a real or quaternionic scalar of modulus one. Each choice yields the analogue of Theorem 8.3 for the corresponding case of Theorem 3.6, and Theorem C holds in all three cases, since the vectors of Table 1 have real coordinates.

Remark 10.4 (The logical form of the axioms). The axioms (H1), (OL), (P1)–(P3), (Sub0), (Sub4), (At1), (At2), (K1), (K2), (K4), (Sp1), (M1) and (M2) are universal conditions built from finitary formulas. Axioms (H2), (Sub2), (Sub3), (K3), (Sp2), (M3) and the formula θ use countable connectives whose arguments are uniformly bounded. Axioms (P4), (Sub1), (At4), (At5), (Sp2), (Sp3) and (M3) use the device (2), which needs only uniform bounds; the functions at, rep and span are not continuous, as Section 5 allows. Axiom (At3) uses the formula $\chi(a)$, which contains a quantifier over the Solèr sort.

Remark 10.5 (Redundancy). In standard models, several axioms follow from others: (Sub1) from (Sub2) and (P4); (Sub3) from (Sub2), (Sub4), (OL) and (P4); (At3) from (At1), (At2) and Proposition 7.3; and (At4) from (At1), (P4) and Proposition 7.3. These axioms are stated nonetheless, so that each property is visible in the axioms.

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